STUDENT PERFORMANCE ANALYSIS PROJECT REPORT

Project Title	Student Performance Analysis
Course / Section	PROGRAMMING IN PYTHON [B]
Instructor	MD. TANZEEM RAHAT
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Submitted By:

Name	**ID**
BADHON KUMAR BISWAS	22-47992-2
DIAMOND HALDER	22-48152-2
ANTARA SAHA	22-48052-2

1. Executive Summary

This project analyzes a dataset containing the daily habits, academic performance, and demographic details of university students. The primary goal is to investigate how lifestyle factors specifically study hours, screen time, and sleep that impact final exam scores. The analysis involved comprehensive data cleaning, including duplicate removal and missing value imputation, alongside outlier detection using the Interquartile Range (IQR) method. Custom features were engineered to evaluate study efficiency and total digital distraction. Key findings reveal that study hours have the strongest positive correlation with exam scores, while excessive screen time significantly deteriorates performance even among students who study frequently. Additionally, optimal sleep paired with high study hours yields the highest academic outcomes.

2. Introduction and Project Objectives

Item	Your Content
Dataset name	Student Performance and Lifestyle Dataset
Dataset source	https://www.kaggle.com/datasets/arooj56/students-habits-performance
Main project goal	To analyze the impact of daily lifestyle habits such as study hours, digital media consumption, and sleep on student's academic performance.
Research Question 1	Do students with higher study time but also high screen time perform better than students with moderate study time and low screen time?
Research Question 2	What is the optimal combination of sleep and study hours that leads to maximum academic performance?
Research Question 3	Among study hours, mental health, and attendance, which factor has the strongest influence on exam scores?

3. Dataset Description

The dataset contains 1,020 student records across 16 variables capturing demographics, study behavior, digital entertainment usage, physical health indicators, and academic outcomes. Each row represents a single student's self-reported lifestyle metrics alongside a measured exam score. The target variable exam_score ranges from 18.4 to 100.0 with a mean of 69.6.

3.1 Basic Dataset Information

Property	Value
Number of rows	1020
Number of columns	16
Data types present	Numerical, Categorical

Target or key variable(s)	exam_score
Potential issues observed at first glance	Missing values, duplicates, outliers

3.2 Initial Observations

An initial inspection revealed that the dataset includes exactly 20 duplicate rows that could skew aggregated metrics. The column names contain inconsistent spacing and capitalization, which requires standardization for programmatic access. Furthermore, there are exactly 92 missing values isolated entirely within the parental_education_level column.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Duplicate records	Dropped 20 duplicate rows	Duplicate entries can skew aggregation results and falsely inflate statistical significance.	Step 1: Duplicate Handling --- Cell [4]
Inconsistent column names	Converted to lowercase and replaced spaces with underscores	Standardizing column names prevents syntax errors and ensures seamless access to DataFrame series.	Step 2: Column Standardization --Cell [4]
Missing values in parental_education_level	Filled 92 null values with the column's mode	Since the data is categorical, imputing with the most frequent value is a standard method to retain dataset volume without distorting the distribution.	Step 3: Missing Value Handling --Cell [4]
Statistical outliers in numeric variables	Identified outliers using the Interquartile Range (IQR) method	Identifying outliers ensures that extreme values (e.g., exam score of 18.4) are recognized when interpreting data distributions.	Part 1: IQR Outlier Calculation ---Cell [5]

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
study_category	Grouped study_hours_per_day into "Low" (<2), "Medium" (2-5), and "High" (>5)	Allows for categorical performance grouping based on student effort levels.
total_screen_time	Sum of social_media_hours and netflix_hours.	Provides a single unified metric to assess digital distraction and screen consumption.
study_efficiency	exam_score divided by (study_hours_per_day + 0.01).	Provides a ratio of academic output per hour of study effort.
entertainment_ratio	total_screen_time divided by (study_hours_per_day + 0.01).	contrasting entertainment time against productive study time.
sleep_category	Grouped sleep_hours into "Low" (<6), "Normal" (6-8), and "High" (>8).	Categorizes rest patterns to evaluate the impact of sleep deprivation or oversleeping on performance.

6. Analysis Methodology

- **Pandas** was utilized for loading the CSV file, dropping duplicates, imputing missing values, and engineering categorical bins and ratio-based features.
- **NumPy** was utilized for numeric outlier computation across all quantitative columns utilizing the 25th and 75th percentiles to establish boundaries.
- **Matplotlib and Seaborn** were used to generate subplots, bar charts, and a correlation heatmap to visually answer the research questions.

7. Analysis and Visualizations

7.1 Research Question 1

Do students with higher study time but also high screen time perform better than students with moderate study time and low screen time?

To answer this, students were grouped by their study_category combined with a binned total_screen_time attribute. The mean exam scores for these specific demographic intersections were then computed.

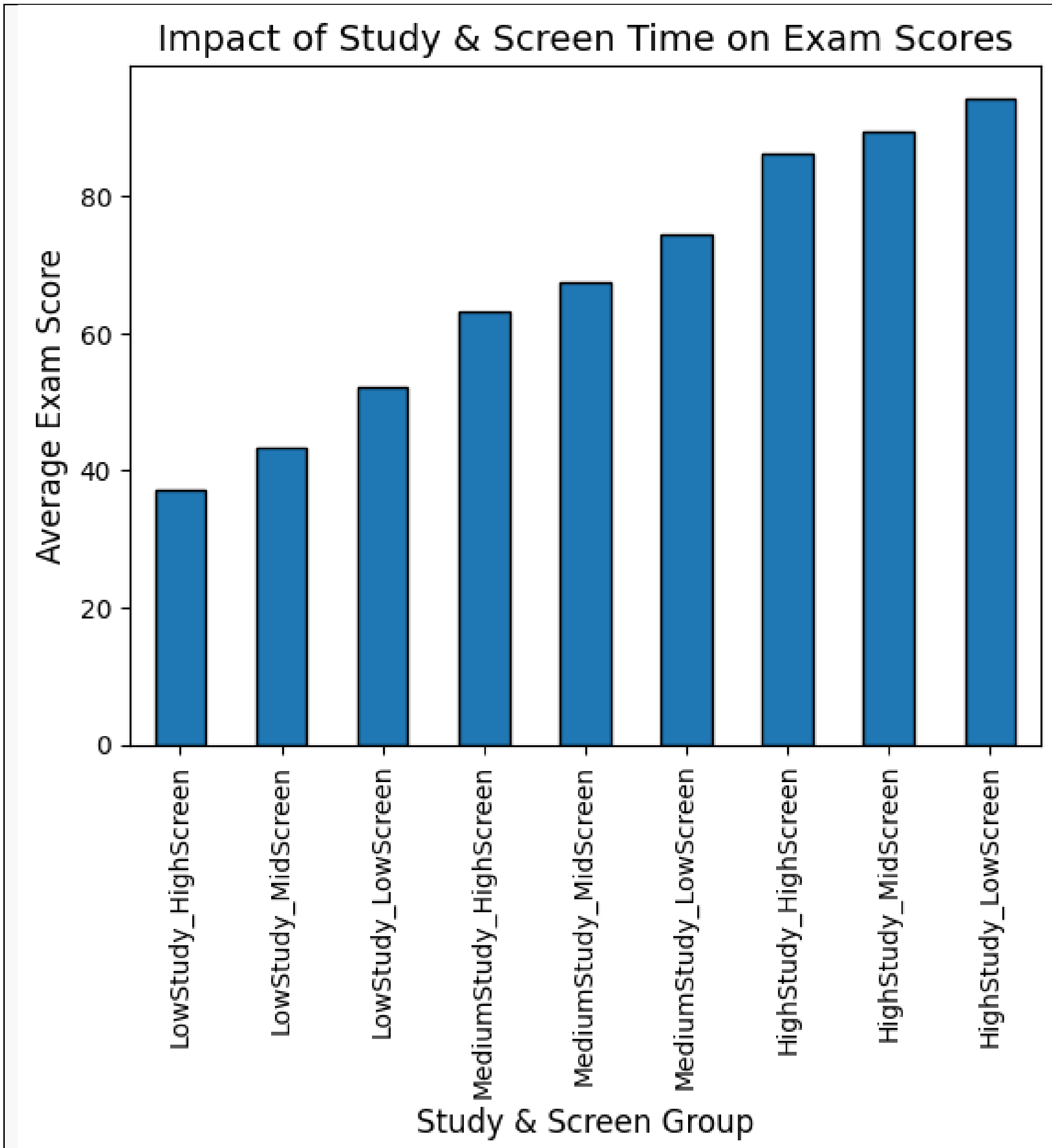

Figure 1. Impact of Study & Screen Time on Exam Scores.

Students categorized as "HighStudy_LowScreen" achieved the highest average scores at 94.41. Conversely, students in the "LowStudy_HighScreen" group scored the lowest, averaging only 37.06. The data indicates that while high study hours predictably raise scores, excessive screen time actively penalizes performance within every study category.

7.2 Research Question 2

What is the optimal combination of sleep and study hours that leads to maximum academic performance?

The data was grouped using the engineered sleep_category and study_category variables, and the mean exam_score for every possible combination was calculated and visualized

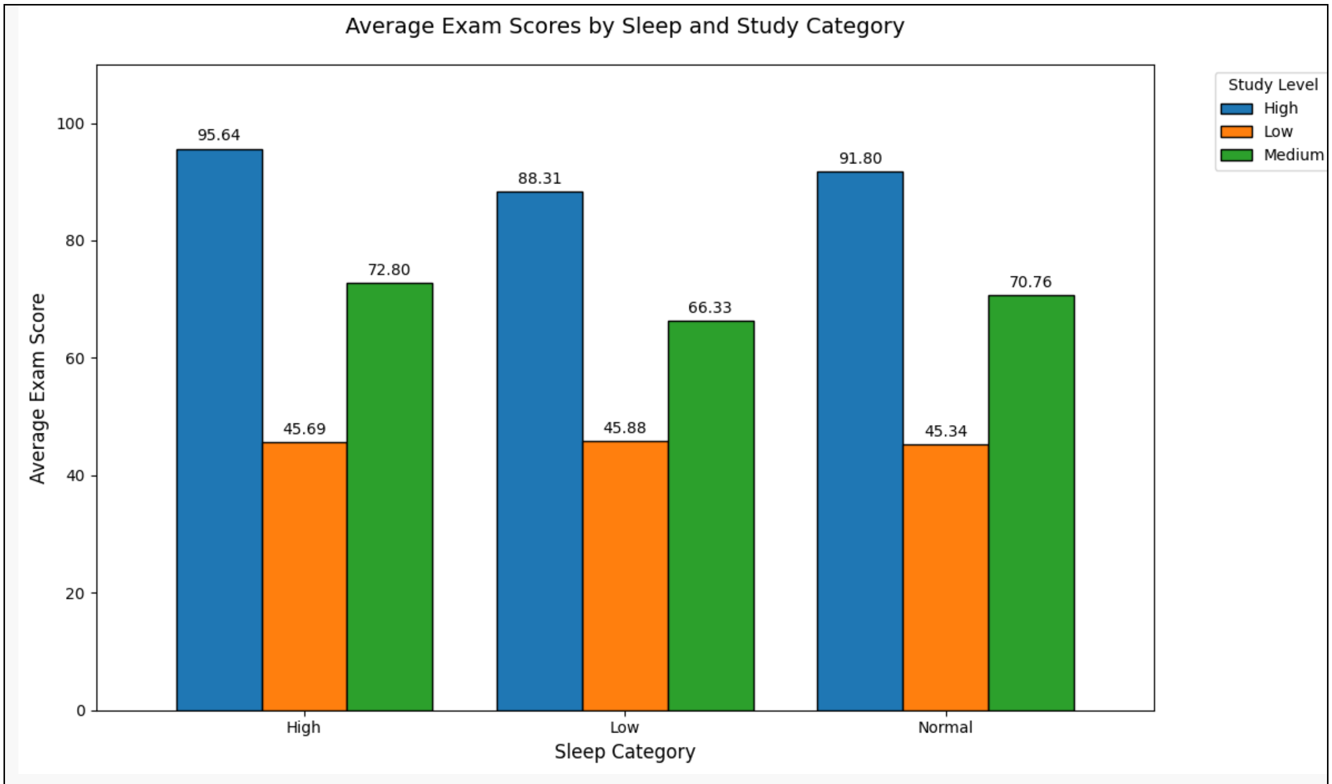

Figure 2. Average Exam Scores by Sleep and Study Category.

Students who maintain high study hours and get "High" sleep (>8 hours) score the highest overall, averaging 95.64. A lack of adequate rest ("Low" sleep category) drops the average score for high-studying students down to 88.31. Across all sleep levels, those who study "Low" hours consistently score poorly (averaging roughly 45), indicating that good sleep cannot compensate for a lack of studying

7.3 Research Question 3

Among study hours, mental health, and attendance, which factor has the strongest influence on exam scores?

A Pearson correlation matrix was generated strictly for the numerical columns to evaluate the linear relationships between the independent variables and the target variable (exam_score).

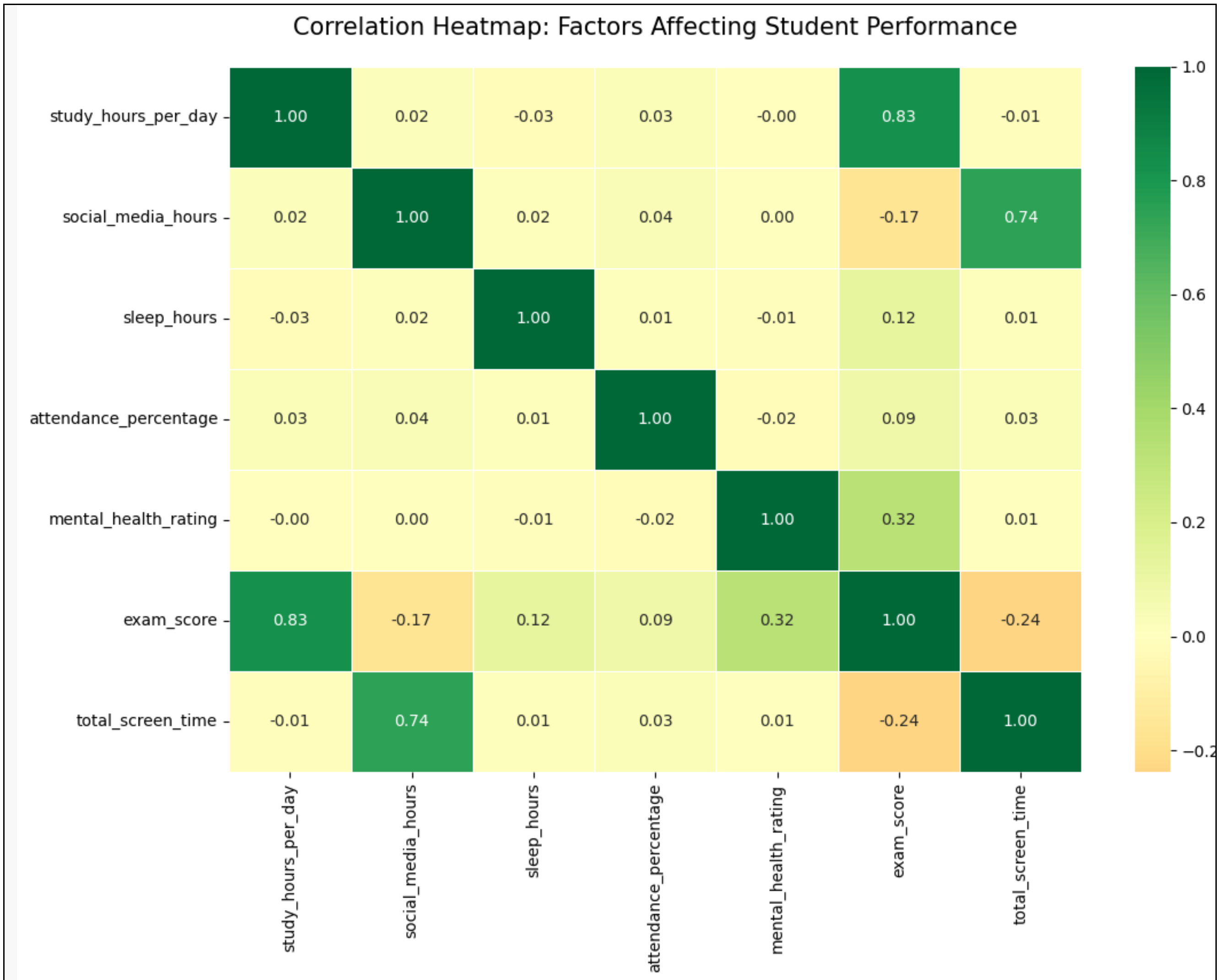

Figure 3. Correlation Heatmap: Factors Affecting Student Performance.

Study hours per day demonstrated the strongest positive correlation with exam scores at 0.83. Mental health rating had a moderate positive correlation of 0.32, while attendance percentage showed a surprisingly weak correlation of only 0.09. Therefore, dedicated study time is the overwhelming driving factor behind exam success.

Outlier Detection — Boxplots

IQR-based outlier detection was applied to all 9 numerical variables. The multi-panel boxplot below visualizes the distribution and any extreme values across each variable simultaneously.

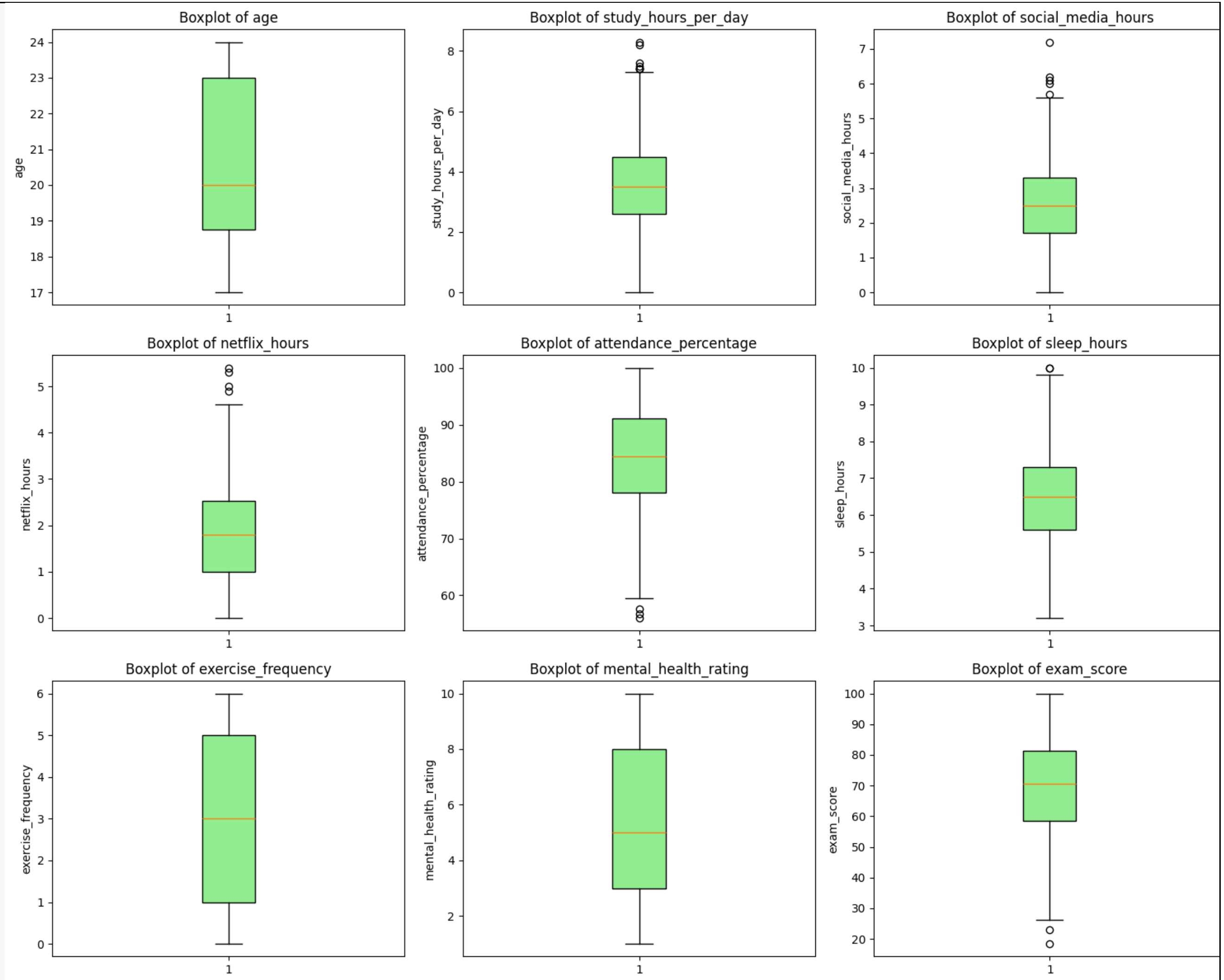

Figure 4. Multi-panel boxplot displaying the distribution and potential outliers across nine continuous and discrete numeric variables.

The boxplots show the normal range of the data and help us find unusual values (outliers). Variables like age, exercise frequency, and mental health do not have any outliers. However, a few students spend an unusually high amount of time on studying, social media, and Netflix (shown by the dots at the top). On the other hand, a small group of students has very low attendance and exam scores (shown by the dots at the bottom). This means a few students have very different habits and academic results compared to the rest of the class.

8. NumPy-Based Custom Computation

A custom Learning Efficiency Score was computed using NumPy's vectorized np.divide() function, measuring how many exam marks each student earned per hour of study invested.

Component	Description
Computation used	division using np.divide() and mean calculate using np.mean()
Purpose	Identify students achieving high scores with relatively fewer hours — highlighting learning efficiency vs. raw effort
Formula or logic	efficiency = np.divide(score_arr, (study_arr + 1)) np.mean(efficiency).round(2)
Result / takeaway	16.22 marks per study hour across the dataset

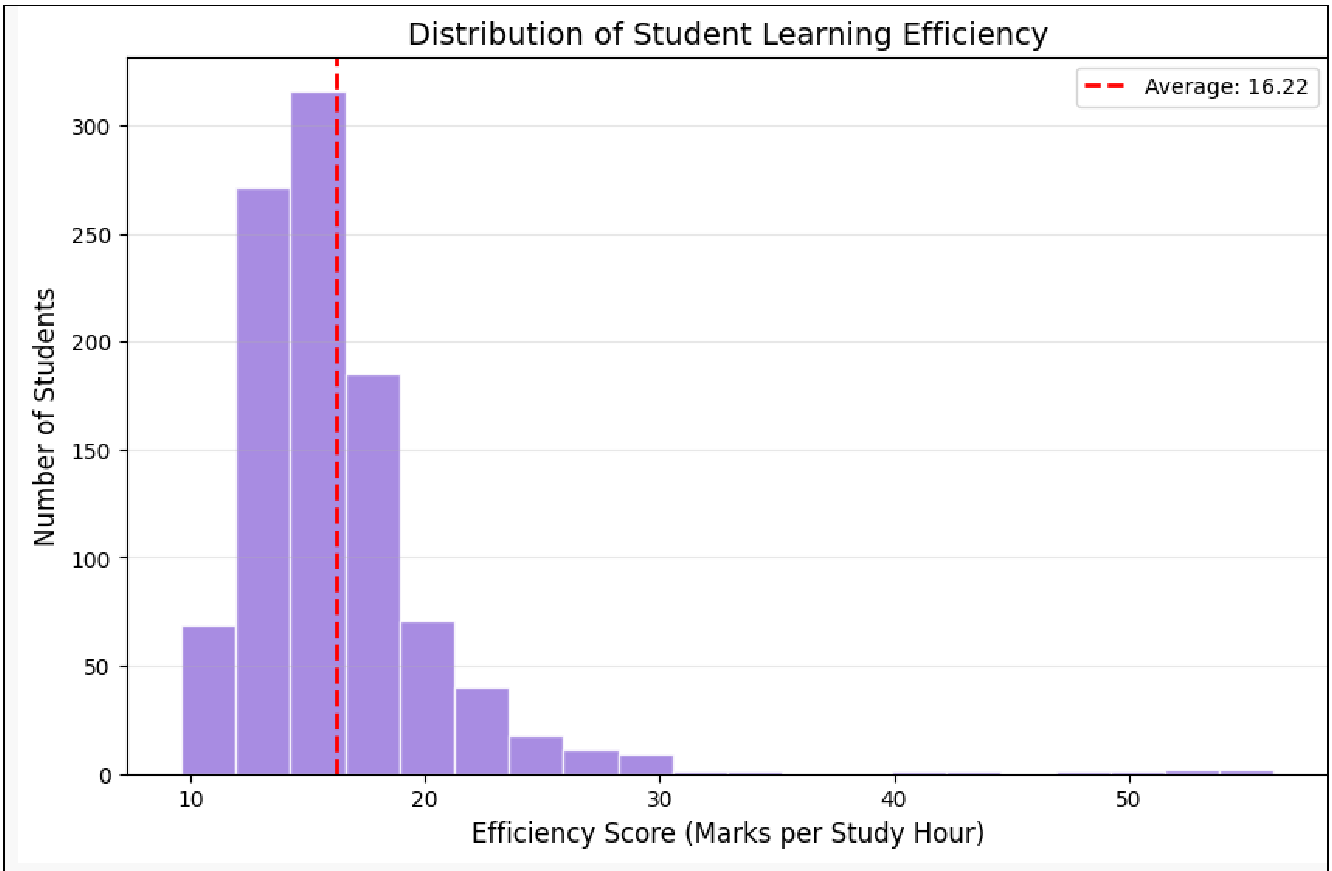

Figure 5. Distribution of Student Learning Efficiency

This graph shows how efficiently students learn by calculating the marks they get for every single hour they study. Most students are clustered together on the left side, earning between 10 and 20 marks for each hour of study. The red dashed line shows that the overall average is 16.22 marks per hour.

Why we used NumPy: Instead of doing the math one by one (like a slow Python loop), NumPy divides the numbers for all 1,000 rows instantly at the exact same time. This makes the code run much faster.

Why we added +1 (or +0.01): We added a small number to the study hours before dividing. This is a common coding trick to stop the program from crashing (a "divide-by-zero" error) just in case a student reported exactly 0 study hours.

9. Key Findings

1. **Studying is the Most Important Thing:** The amount of time you spend studying has the strongest link (0.83) to your final score. More study time directly leads to better grades.
2. **Too Much Screen Time Hurts Grades:** Spending a lot of time on screens lowers exam scores (-0.24). Even if a student studies hard, too much screen time will still bring their marks down.
3. **Good Sleep Makes Hard Work Pay Off:** The students who get the absolute highest scores (averaging 95.64) are the ones who both study a lot AND get plenty of sleep.
4. **Mental Health Matters:** Students with better mental health tend to get better scores (0.32). Feeling good mentally helps you perform well in exams.
5. **Just Attending Class Isn't Enough:** Surprisingly, just being present in class has a very weak link (0.09) to getting a high exam score.
6. **There Are No "Magic" High Scores:** We checked to see if any students in the "Low Study" group got unusually high exam scores, and we found exactly 0 students. This proves that no one can get a top grade without actually putting in the time to study.

10. Limitations

This project has a few limitations that we should keep in mind:

- **Self-Reporting Guesswork:** The data relies on students reporting their own habits. Often, people guess their daily hours or hide their bad habits (like under-reporting screen time or over-reporting study time). This means the numbers might not be 100% perfect.
- **Correlation is not Causation:** We found a strong link between high study hours and high exam scores. However, we cannot say studying is the *only* reason for success. Other important things are missing from this dataset, like the student's natural talent, the quality of their teachers, or their family's financial support.
- **Missing Data Fixes:** There were 92 missing values in the "parental education" column. We filled these empty spaces with the most common answer (the mode). While this is a standard data cleaning step, it might not show the true education level of those specific parents.

11. Conclusion

The main objective of this project was to analyze how a student's lifestyle and daily habits affect their academic performance. After exploring the dataset, the most important takeaway is very clear: dedicated study time is the single biggest driver of high exam scores. In fact, putting in a high number of study hours is so effective that it can even overcome the negative effects of spending too much time on digital screens.

However, hard work alone isn't the only secret to success. The analysis revealed that the absolute top-performing students are those who maintain a healthy balance by combining high study hours with a high amount of sleep. We also found that good mental health plays a supportive role in achieving high grades, while simply attending classes isn't enough without putting in the actual study effort.

If more time and data were available, a great next step would be to build a Machine Learning model. This model could predict a student's exact exam score based on their daily routine, which would be a highly valuable tool for teachers and parents to help guide struggling students toward better habits.

12. References / Dataset Source

Dataset Information

- **Dataset Name:** Student Habits & Performance Dataset
- **Primary Source:** Kaggle - Students Habits & Performance
- **Raw Data File:** Google Drive Download Link

Development Environment & Tools

- **Programming Language:** Python 3.14.3
- **Development Environment:** Jupyter Notebook
- **Libraries Utilized:**
    - Pandas - *For data manipulation and aggregation*
    - NumPy - *For numerical and statistical computations*
    - Matplotlib & Seaborn - *For data visualization*